cloudspaces

D2.1: Roadmap of Target Outcomes

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SEVENTH FRAMEWORK PROGRAMME

UNIVERSITAT ROVIRA I VIRGILI

Contents

- Taxonomy of Personal Clouds
- Open Challenges
- Roadmap of Outcomes



From the Personal Computer to the Personal Cloud

The Personal Computer implied that users had absolute control of their device's hardware and software.

But this recent multi-device approach clearly implies access to Cloud resources that may compromise the privacy and control of user's information.



From PC to Personal Cloud



3S Personal Cloud Definition

The Personal Cloud is a unified digital locker for users' personal data offering three key services: **Storage, Synchronization and Sharing.** On the one hand, it must provide redundant and trustworthy cloud data **storage** for users' information flows irrespective of their type. On the other hand, it must provide **syncing** and file exploring capabilities in different heterogeneous platforms and devices. And finally, it must offer fine-grained information **sharing** to third-parties (users and applications).

State of the art

- Review of commercial and open source initiatives regarding:
  - Storage
  - Sync
  - Share
  - Privacy
  - Platforms



Conclusions

- Complete lack of interoperability
- Very few platforms address privacy issues
- Open source alternatives (ownCloud) lack important services:
  - Scalability, reliability
  - Privacy
    - Client-side encryption
    - Privacy-aware sharing
  - Interoperability



Open Challenges

- **Privacy**
  - Privacy Enhancing Technologies
    - Encryption and access control
  - Automated Privacy-Aware Data Sharing
    - Context-aware data sharing
- **Interoperability**
  - Syntactic
    - APIs for Storing and Sharing
  - Semantic

    Semantic-aware data sharing, metadata

Open Challenges

Home Device Personal Clouds

- Insufficient reliability and durability
- Lack genuine elasticity
- Privacy is in friction with availability
- Manageability
- Trend towards hybrid systems

- **Cloud Storage services**
  - Data transfer bottlenecks, content distribution
  - Data optimization and management

Roadmap of Outcomes

A1	Roadmap of target outcomes	✓
A2	Validation platform and deployment of testbed infrastructures	✓
A3	Generation of traces from U1 and eyeOS	✗
A4	Early prototype of Adaptive Personal Storage	✓
A5	Early prototype of Adaptive Cloud Storage	✓
A6	Early prototype of Privacy-aware Sharing service	✓
A7	Design of Syntactic APIs	✓

Roadmap of Outcomes

A8 Persistence service

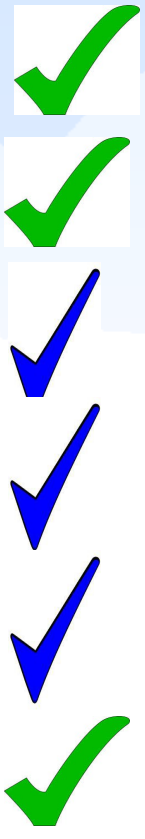
A9 eyeFiles connection with Store API

A10 Design of secure sharing with ACLs

A11 Semantic ontologies for Personal Clouds

A12 Proximity-aware Service Design

A13 Early implementation of Store and Share APIs



Roadmap of Outcomes

A14 eyeCalendar integration with U1DB 

A15 Public Release of StackSync Cloud testbed with real users 

A16 - A22 First year Deliverables 

Roadmap of Outcomes

B1	Public release of Privacy-aware sharing in StackSync
B2	Public release of HPDC Adaptive Storage service
B3	Public release of service platform APIs
B4	Public release of eyeOS collaboration tools and open testbed
B5	Launch of User communities on top of open testbeds

Roadmap of Outcomes

B6-B10	Year 2 Deliverables
B11	International Workshop with Personal Cloud Providers

Software Releases

StackSync scheduled for M12-M14
- stacksync.org, stacksync.org, github
- More dissemination activities are planned (OpenStack Summit)

U1DB Web scheduled for M9-10

We plan next releases from M18-20 (U1, eyeOs, StackSync)